

RIPHAH INTERNATIONAL UNIVERSITY

Faculty of Computing
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DATABASE MANAGEMENT SYSTEM PROJECT TRUCKING MANAGEMENT SYSTEM

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Department	BS Cyber Security
Semester	4th Semester
Course	Database Management System
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. Acknowledgement

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. Abstract

The Trucking Management System is a desktop-based application developed using C# programming language with Windows Forms interface in Visual Studio IDE, connected to a SQL Server database for data storage and management. This project was developed as part of the Database Management System (DBMS) course in the 4th Semester of the BS Cyber Security program at Riphah International University.

The main purpose of this system is to provide a centralized and organized platform for managing trucking operations. It handles key aspects such as driver registration, truck registration, shipment management, and user login/authentication. The system allows administrators and users to add, view, search, and manage records related to drivers, trucks, and shipments through a simple and easy-to-use graphical interface.

The system is built using a client-server architecture where the front-end is developed with Windows Forms (WinForms) in C# and the back-end uses SQL Server as the relational database management system. All data is stored in structured SQL tables with proper primary keys and relationships.

This documentation covers all aspects of the project including system design, database structure, form descriptions, working procedures, features, advantages, limitations, and future enhancements. The goal of this project was to apply the theoretical knowledge of DBMS practically and develop a functional software application that solves a real-world problem in the logistics and transportation industry.

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1. Project Overview

The Trucking Management System is a comprehensive desktop application designed to digitalize and simplify the daily operations involved in trucking and logistics management. In today's fast-paced world, transportation and logistics companies need efficient digital tools to manage their fleets, drivers, and shipments without relying on manual paperwork.

This project provides a complete solution for managing trucking operations through a user-friendly Windows Forms application that is connected to a robust SQL Server database. The system allows both admin and regular users to log in securely and perform various operations like adding drivers, managing trucks, handling shipments, and searching records with ease.

1.1 Project Summary

Project Title	Trucking Management System
Developer	Raja Ali Shahid
SAP ID	63158
Department	BS Cyber Security
Semester	4th Semester
University	Riphah International University
Programming Language	C#
IDE	Visual Studio
UI Framework	Windows Forms (WinForms)
Database	SQL Server
Project Type	Desktop Application (DBMS Project)
Year	2026

2. Introduction

Transportation and logistics are among the most important sectors in the modern economy. Every day, thousands of trucks move goods and materials from one place to another, making transportation companies heavily dependent on organized record-keeping and fleet management.

Before the use of computerized systems, trucking companies used to manage their records manually using paper files and registers. This manual approach was not only time-consuming but also prone to human errors, data loss, and inefficiency. The need for a digital solution became essential as the volume of operations grew.

The Trucking Management System has been developed to address these challenges. It provides a centralized platform where all important data related to drivers, trucks, and shipments can be stored, accessed, and managed efficiently. The system eliminates the need for manual records and provides instant access to data through a simple graphical user interface.

2.1 Purpose of the System

The purpose of this system is to:

- Provide a digital platform for managing trucking operations
- Maintain organized records of drivers and their information
- Track truck registration and availability
- Manage shipment details and their statuses
- Provide secure login authentication for users
- Enable fast searching and viewing of records

2.2 Background

This project was developed as a semester project for the Database Management System (DBMS) course at Riphah International University. The project demonstrates the practical implementation of database design, SQL queries, C# programming, and Windows Forms UI development. The student has applied concepts learned in class to build a fully functional application that mimics real-world trucking management operations.

3. Problem Statement

Transportation and trucking companies face significant challenges when managing their operations manually. Some of the major problems include:

1. **Lack of Centralized Data Management:** There is no single organized platform to store all information about drivers, trucks, and shipments. Data is scattered across multiple files and notebooks, making it hard to access when needed.
2. **Time-Consuming Manual Processes:** Adding, updating, or searching for records manually takes a lot of time and effort. This reduces the overall efficiency of the organization.
3. **Data Loss and Inaccuracy:** Paper-based records are prone to loss, damage, and errors. A single mistake in a manual record can cause serious problems in operations.
4. **No Search and Filter Functionality:** Without a computerized system, finding specific records (like a particular shipment or driver) requires manually going through all files, which is very slow.
5. **No User Authentication:** Manual systems do not provide any security mechanism to restrict unauthorized access to sensitive data.
6. **Difficulty in Generating Reports:** Creating reports about shipments, drivers, and truck usage from manual records requires significant effort and time.

The Trucking Management System is developed to solve all these problems by providing a digital, organized, and efficient platform that automates the record-keeping and management of trucking operations.

4. Objectives

The main objectives of developing the Trucking Management System are as follows:

- To develop a fully functional desktop application for trucking management using C# and Windows Forms.
- To design and implement a well-structured SQL Server database for storing all application data.
- To implement a secure user login and authentication system.
- To provide functionality for adding, viewing, and managing driver records.
- To provide functionality for adding, viewing, and managing truck records.
- To develop a complete shipment management module with search functionality.
- To implement CRUD (Create, Read, Update, Delete) operations for all modules.
- To connect the C# application with the SQL Server database using ADO.NET for data connectivity.
- To create an intuitive and easy-to-use graphical user interface suitable for non-technical users.
- To apply concepts of relational database design including primary keys, tables, and SQL queries.
- To demonstrate the practical application of DBMS concepts learned during the 4th Semester.

5. Scope of the System

The Trucking Management System is designed for small to medium-sized trucking companies or educational demonstration purposes. The scope of the system includes:

5.1 In Scope

- User Login and Authentication System
- Driver Registration and Management Module
- Truck Registration and Management Module
- Shipment Creation and Management Module
- View and Search Functionality for all records
- SQL Server database integration with CRUD operations
- Windows-based desktop application

5.2 Out of Scope

- Mobile application version
- Web-based interface
- GPS tracking of trucks
- Online payment integration
- Multi-branch or multi-company support
- Real-time tracking and notifications
- Automatic report generation and export to PDF

6. Technical Skills Used

This project required the use of multiple technical skills and technologies. Below is a summary of the key skills applied during development:

Skill / Technology	Category	Usage in Project
C#	Programming Language	Core application logic, event handling, form operations
Windows Forms	UI Framework	Designing all GUI forms and user interface elements
SQL Server	Database	Storing and managing all application data
Visual Studio	IDE	Development environment for coding and debugging
ADO.NET	Database Connectivity	Connecting C# application to SQL Server
SQL Queries	Database Operations	SELECT, INSERT, UPDATE, DELETE operations
CRUD Operations	Data Management	Create, Read, Update, Delete for all modules

7. System Requirements

7.1 Software Requirements

Software	Details
Operating System	Windows 10 / Windows 11
IDE	Microsoft Visual Studio 2019 / 2022
Programming Language	C# (.NET Framework)
Database	Microsoft SQL Server 2019 / 2022
Database Tool	SQL Server Management Studio (SSMS)
.NET Framework	.NET Framework 4.7.2 or later
UI Framework	Windows Forms (WinForms)

7.2 Hardware Requirements

Component	Minimum Requirement
Processor	Intel Core i3 or equivalent (1.8 GHz or faster)
RAM	4 GB or higher
Storage	20 GB free disk space
Display	1024 x 768 resolution or higher
Input Devices	Keyboard and Mouse
Internet	Not required (offline desktop application)

8. System Architecture

The Trucking Management System follows a two-tier client-server architecture. In this model, the application is divided into two layers:

8.1 Presentation Layer (Front-End)

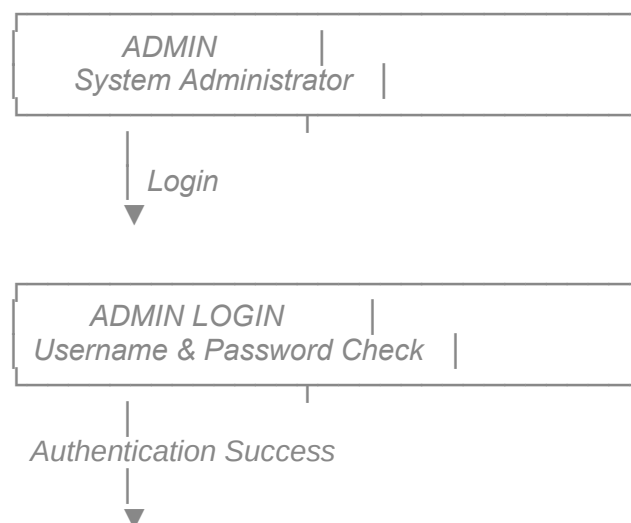
The front-end of the application is built using Windows Forms (WinForms) in C# within Visual Studio. This layer includes all the graphical user interface components such as forms, buttons, text boxes, labels, and data grids. The user interacts directly with this layer to perform operations.

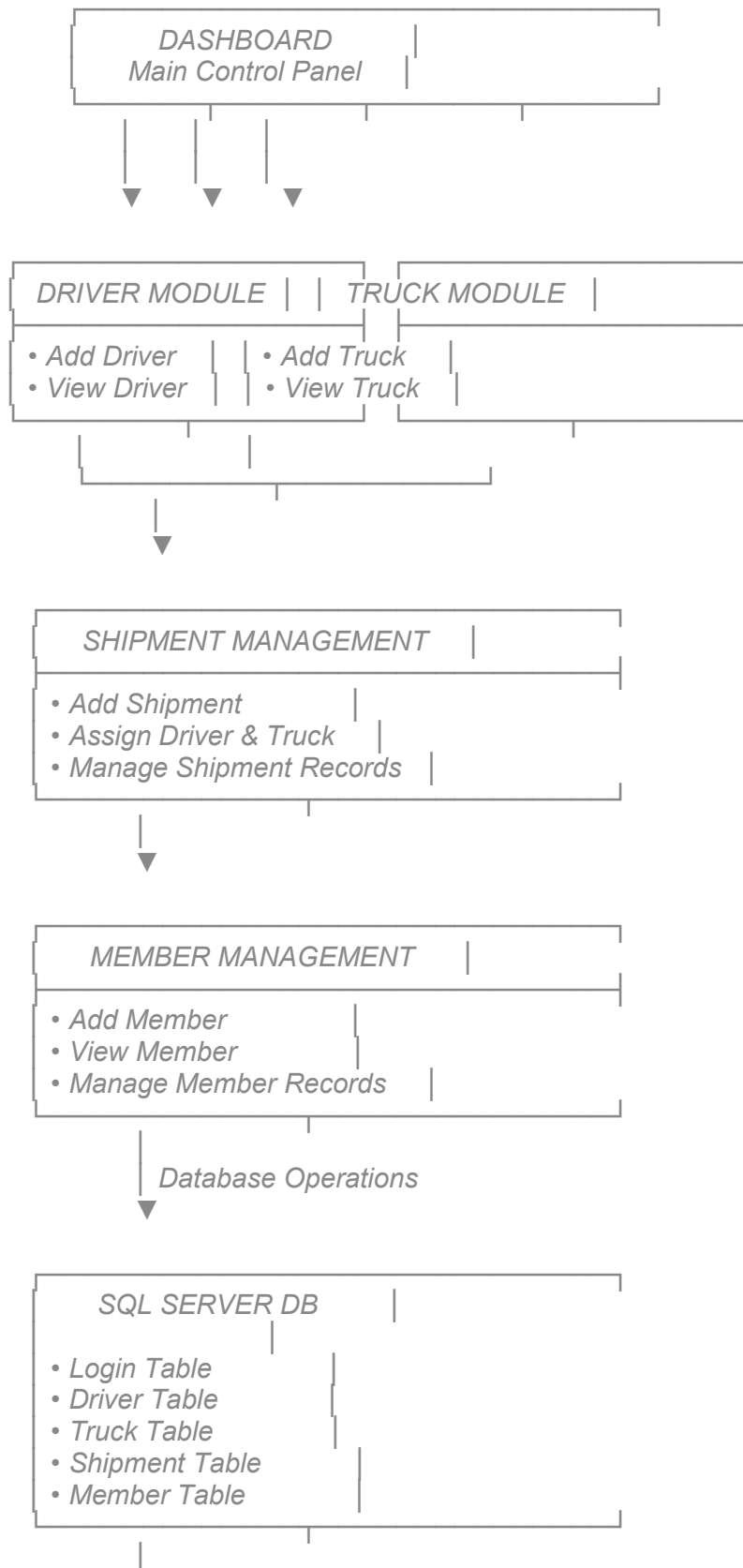
8.2 Data Layer (Back-End)

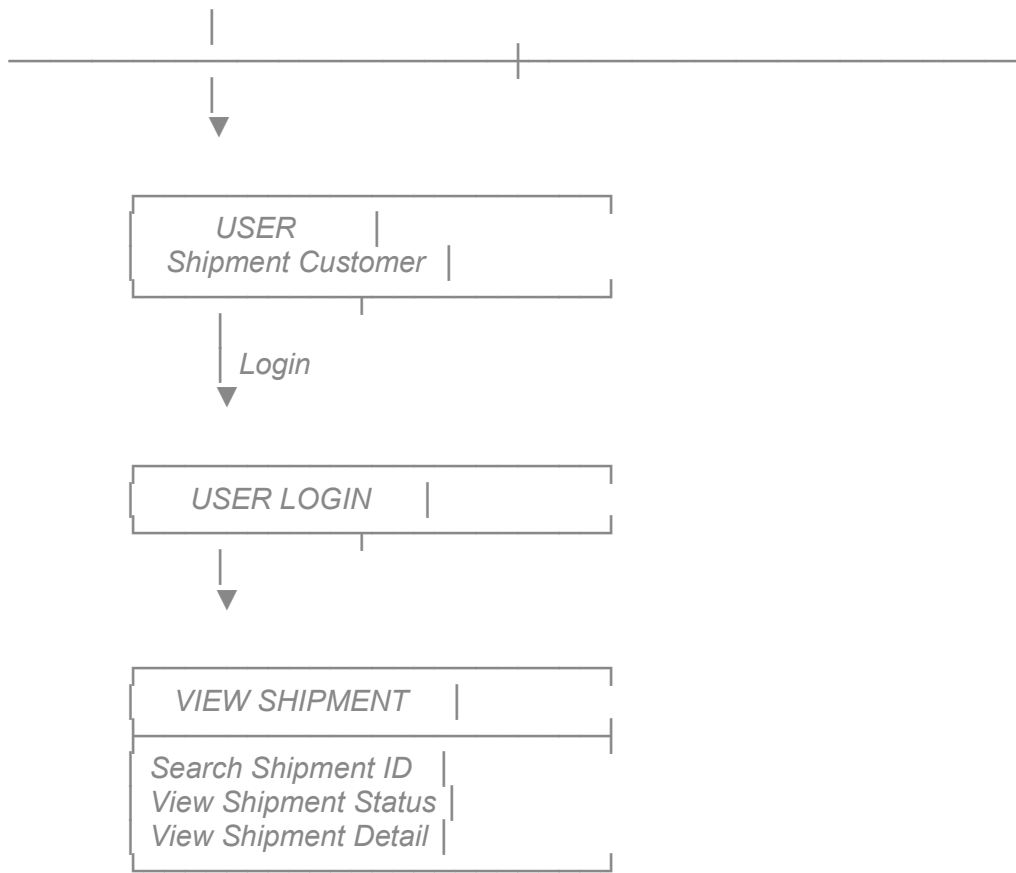
The back-end consists of the SQL Server database which stores all the application data in structured tables. The communication between the front-end and the database is handled through ADO.NET using SqlConnection, SqlCommand, and SqlDataAdapter classes.

Architecture Flow:

- User opens the application and interacts with Windows Forms (Front-End)
- User actions trigger C# event handlers and business logic
- C# code connects to SQL Server via ADO.NET connection string
- SQL queries are executed to fetch, insert, update, or delete data
- Results are returned to the front-end and displayed to the user







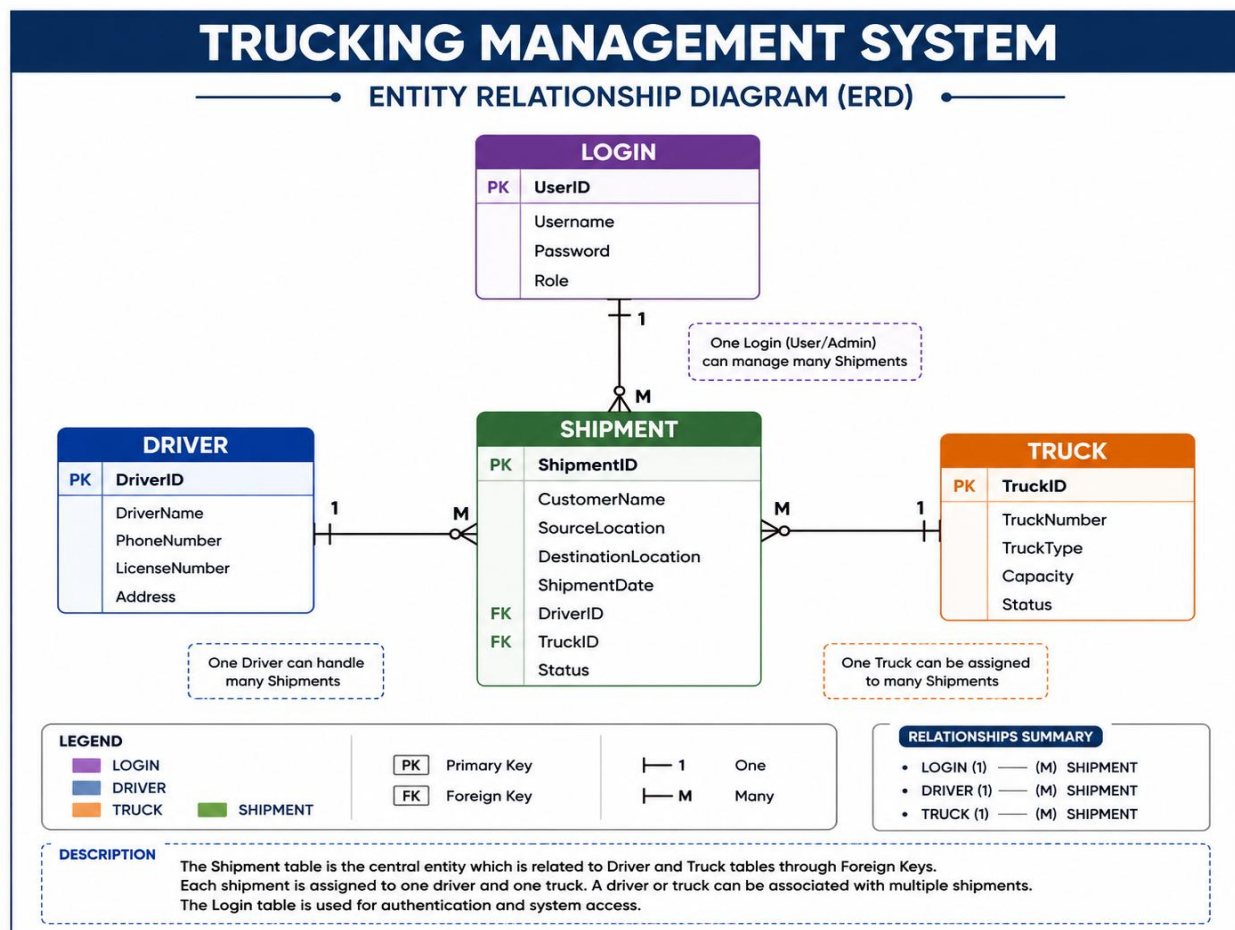
9. Database Design

The database for the Trucking Management System is designed using Microsoft SQL Server. It follows a relational database model with multiple tables that store different types of data. Each table has a primary key and is related to other tables where necessary.

9.1 Database Overview

Database Name	TruckingManagementDB
DBMS Used	Microsoft SQL Server
Number of Tables	4 (Login, Driver, Truck, Shipment)
Design Type	Relational Database
Connection Method	ADO.NET (SqlConnection)

9.2 ER Diagram



9.3 Table Relationships

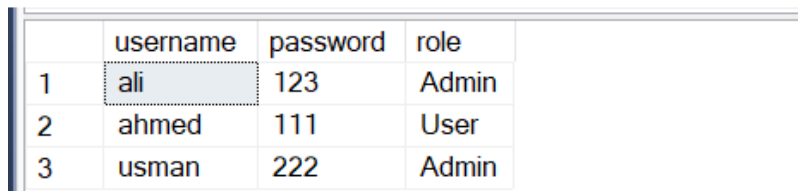
The database consists of four main tables. The Login table manages user authentication. The Driver table stores driver information. The Truck table stores truck details. The Shipment table stores shipment records. These tables work together to provide a complete trucking management solution.

10. SQL Tables Explanation

The Trucking Management System database consists of four main SQL tables. Each table is described in detail below along with its screenshot, columns, primary key, and purpose.

10.1 Login Table

The Login table is responsible for storing user credentials for authentication. It ensures that only authorized users can access the system. Each user has a unique username and a corresponding password.



	username	password	role
1	ali	123	Admin
2	ahmed	111	User
3	usman	222	Admin

Figure 10.1 — Login Table (SQL Server)

Column Name	Data Type	Constraint	Description
UserID	INT	PRIMARY KEY	Unique identifier for each user
Username	VARCHAR(50)	NOT NULL, UNIQUE	Login username of the user
Password	VARCHAR(50)	NOT NULL	Password for login authentication
Role	VARCHAR(20)	DEFAULT 'User'	User role (Admin / User)

10.2 Available Driver Table

The Available Driver table stores all information related to drivers registered in the system. When a new driver is added through the Add Driver form, the data is saved in this table. The table contains the driver's personal and professional details.

	driver_id	driver_name	phone	license_number	status
1	1	Ali	03001234567	LIC123	Available
2	2	Ahmed	03111234567	LIC456	Busy
3	3	Usman	03221234567	LIC789	Available

Figure 10.2 — Available Driver Table (SQL Server)

Column Name	Data Type	Constraint	Description
DriverID	INT	PRIMARY KEY	Unique identifier for each driver
DriverName	VARCHAR(100)	NOT NULL	Full name of the driver
LicenseNo	VARCHAR(50)	UNIQUE	Driver's license number
PhoneNo	VARCHAR(20)	NOT NULL	Driver's contact number
Status	VARCHAR(20)	DEFAULT 'Available'	Current availability status

10.3 Available Truck Table

The Available Truck table stores all truck-related information. When a new truck is registered through the Add Truck form, its details are saved in this table. This table tracks all trucks along with their registration numbers and current status.

	truck_number	model	capacity	status
1	4156	2010	15	Active
2	7821	2018	20	Maintenance
3	9001	2022	25	Active

Figure 10.3 — Available Truck Table (SQL Server)

Column Name	Data Type	Constraint	Description
TruckID	INT	PRIMARY KEY	Unique identifier for each truck
TruckNo	VARCHAR(50)	NOT NULL, UNIQUE	Registration number of the truck
Model	VARCHAR(100)	NOT NULL	Model/type of the truck
Capacity	VARCHAR(50)	NOT NULL	Load capacity of the truck
Status	VARCHAR(20)	DEFAULT 'Available'	Current availability status

10.4 Shipment Management Table

The Shipment Management table is the most important table in the system. It stores all details related to shipments including the origin, destination, assigned driver, assigned truck, and current status. Each shipment is identified by a unique shipment ID.

	shipment_id	truck_id	driver_id	shipment_from	destination	shipment_date	shipment_status
1	104	4156	Ali	Muzaffarabad	Pindi	2026-05-09	In Transit
2	105	7821	Ahmed	Muzaffarabad	Haripur	2026-05-10	Pending
3	106	9001	Usman	Muzaffarabad	Neelum ...	2026-05-11	Delivered

Figure 10.4 — Shipment Management Table (SQL Server)

Column Name	Data Type	Constraint	Description
ShipmentID	INT	PRIMARY KEY	Unique identifier for each shipment
Origin	VARCHAR(100)	NOT NULL	Starting location of the shipment
Destination	VARCHAR(100)	NOT NULL	Delivery location of the shipment
DriverID	INT	FOREIGN KEY	Assigned driver for the shipment
TruckID	INT	FOREIGN KEY	Assigned truck for the shipment
Status	VARCHAR(30)	DEFAULT 'Pending'	Current status of the shipment
Date	DATE	NOT NULL	Date of the shipment

11. Forms Design and Explanation

The Trucking Management System consists of multiple Windows Forms screens, each designed for a specific purpose. All forms are built using C# Windows Forms in Visual Studio. Below is the detailed explanation of each form along with its screenshot.

11.1 Main Dashboard

The Main Dashboard is the central hub of the Trucking Management System. It provides access to all major modules of the application through a clean and organized interface. The dashboard is displayed after successful login and shows navigation options for all system features.



Figure 11.1 — Main Application Dashboard

Dashboard Features:

- Central navigation to all modules
- Quick access buttons for Driver, Truck, and Shipment management
- Clean and professional layout
- User-friendly interface suitable for non-technical users

11.2 Login Form

The Login Form is the first screen that appears when the application is launched. It requires the user to enter valid credentials (username and password) to access the system. This form provides security and ensures that only authorized personnel can use the application.



Figure 11.2 — User Login Form

Login Form Components:

- Username Text Box: The user enters their registered username here
- Password Text Box: The user enters their password (input is hidden for security)
- Login Button: Validates the credentials against the Login table in SQL Server
- Error Message: Displays 'Invalid credentials' if login fails

Working Process:

7. User opens the application
8. Login form appears on the screen
9. User enters username and password
10. System queries the Login table in SQL Server
11. If credentials match: user is redirected to the Dashboard
12. If credentials do not match: error message is displayed

11.3 Add Driver Form

The Add Driver Form allows the administrator to register new drivers into the system. All driver information is collected through this form and stored in the Available Driver table in the SQL Server database.

Figure 11.3 — Add Driver Form

Add Driver Form Components:

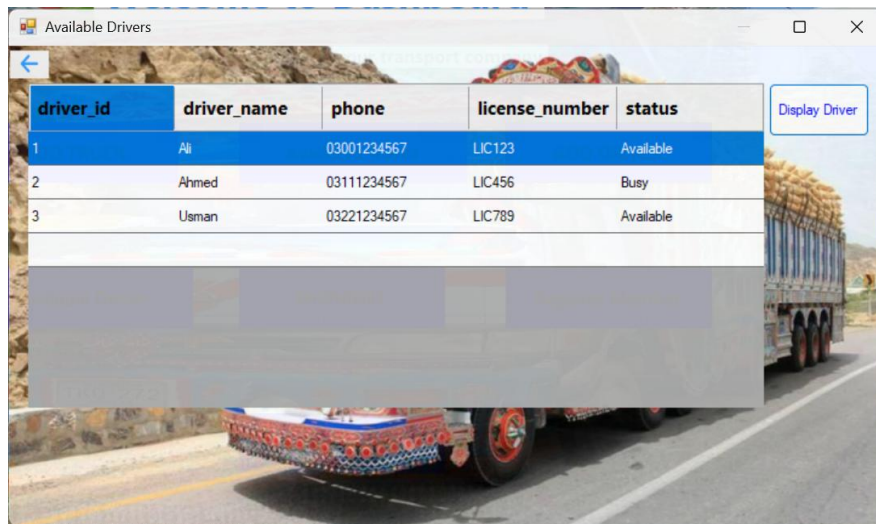
- Driver Name Field: Input field for the full name of the driver
- License Number Field: Input for the driver's license number
- Phone Number Field: Contact number of the driver
- Status Dropdown: Driver availability status (Available / Assigned)
- Add / Save Button: Saves the driver record to the database
- Clear Button: Clears all input fields

Working Process:

13. Admin opens the Add Driver form from the Dashboard
14. Admin fills in all required driver details
15. Admin clicks the Add/Save button
16. System validates the input data
17. SQL INSERT query is executed to save the record
18. Confirmation message is displayed

11.4 View Driver Form

After adding a driver, the system displays all registered drivers in the View Driver form. This form retrieves all records from the Available Driver table and shows them in a DataGridView for easy viewing and management.



The screenshot shows a web application window titled 'Available Drivers'. It features a table with five columns: driver_id, driver_name, phone, license_number, and status. There are three rows of data. A 'Display Driver' button is located to the right of the table. The background of the window shows a truck on a road.

driver_id	driver_name	phone	license_number	status
1	Ali	03001234567	LIC123	Available
2	Ahmed	03111234567	LIC456	Busy
3	Usman	03221234567	LIC789	Available

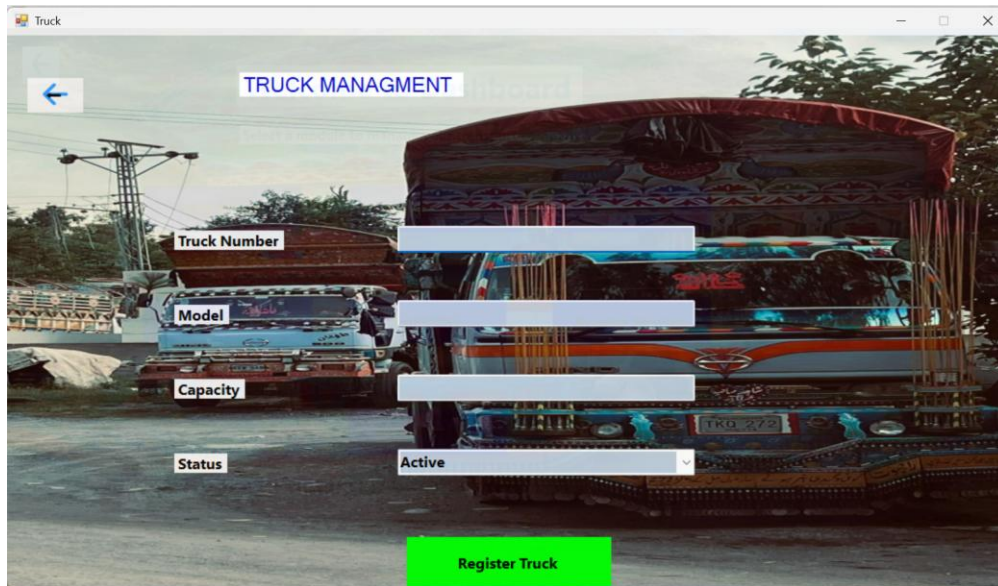
Figure 11.4 — View Driver Records

View Driver Form Features:

- DataGridView: Shows all driver records in a tabular format
- All driver columns are displayed: ID, Name, License, Phone, Status
- Records are loaded automatically from the database when the form opens
- Users can scroll through all driver records

11.5 Add Truck Form

The Add Truck Form allows the registration of new trucks into the system. Truck details including the registration number, model, and capacity are entered through this form and saved to the Available Truck table in the database.



The screenshot shows a web browser window titled 'Truck'. Inside, there's a form titled 'TRUCK MANAGMENT' (note the typo). The form has a back arrow in the top left. It contains four input fields: 'Truck Number', 'Model', 'Capacity', and 'Status'. The 'Status' field is a dropdown menu currently showing 'Active'. At the bottom of the form is a green button labeled 'Register Truck'. The background of the form is a photograph of a large, ornate truck with a red canopy.

Figure 11.5 — Add Truck Form

Add Truck Form Components:

- Truck Registration Number: Input field for the vehicle registration plate
- Truck Model: Input for the model/make of the truck
- Capacity Field: The load capacity of the truck
- Status Dropdown: Truck availability status
- Add / Save Button: Saves truck record to the database
- Clear Button: Resets the form

11.6 View Truck Form

The View Truck Form displays all registered trucks in a data grid. It retrieves records from the Available Truck table and presents them in an organized format for easy viewing.

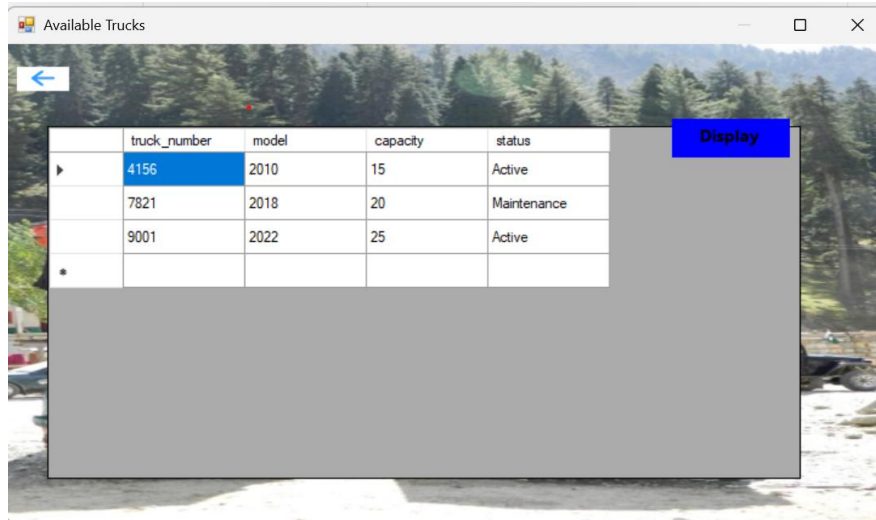


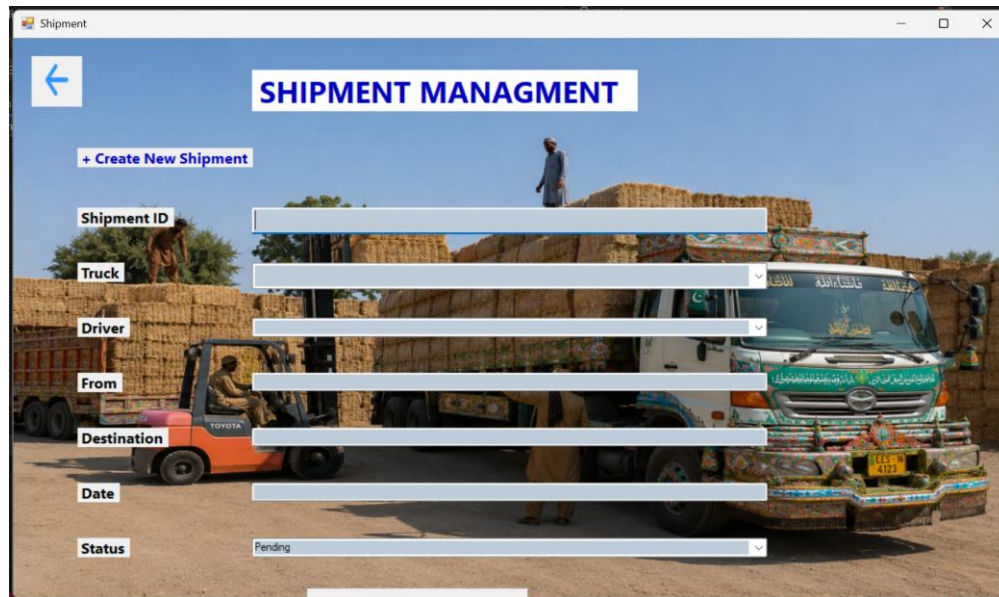
Figure 11.6 — View Truck Records

View Truck Form Features:

- DataGridView showing all truck records
- Columns include: Truck ID, Registration No, Model, Capacity, Status
- All records loaded automatically from SQL Server on form open

11.7 Shipment Management Form

The Shipment Management Form is the main operational form of the system. It allows users to create new shipments, view all existing shipments, and search for specific shipments using a search function. This form connects all the modules together.



The screenshot shows a web application window titled "Shipment". Inside, there's a header with a back arrow and the title "SHIPMENT MANAGEMENT". Below the header is a button "+ Create New Shipment". The form consists of several input fields: "Shipment ID" (text), "Truck" (dropdown), "Driver" (dropdown), "From" (text), "Destination" (text), "Date" (text), and "Status" (dropdown). The "Status" dropdown is currently showing "Pending". The background image shows a truck being loaded with hay by a forklift.

Figure 11.7 — Shipment Management Form

Shipment Management Form Components:

- Origin and Destination Fields: Source and destination locations
- Driver Dropdown: Select from available registered drivers
- Truck Dropdown: Select from available registered trucks
- Shipment Date: Date picker for the shipment date
- Add Shipment Button: Creates a new shipment record
- View All Button: Loads all shipments in the DataGridView
- Search by ID: Allows searching a specific shipment using Shipment ID

11.8 User Login View

This form shows the regular user login interface. Unlike the admin login, the user login has limited access to certain features. Regular users can view shipments and check their status but cannot add or delete records.

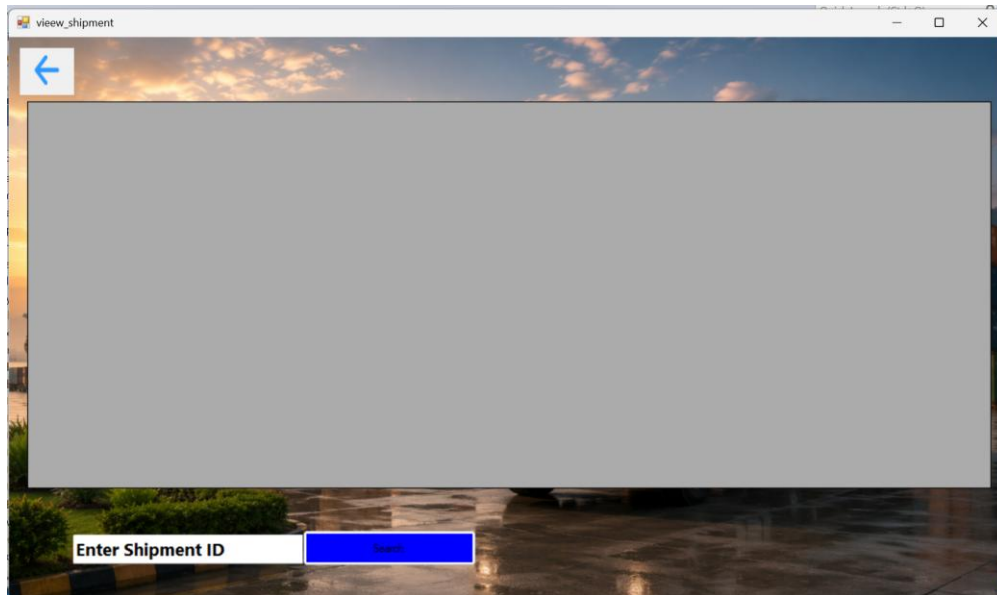


Figure 11.8 — User Login Screen

11.9 After Login — View Shipment

After a successful login, the user is taken to the main interface where they can view all shipments. This screen shows a DataGridView populated with all shipment records retrieved from the Shipment Management table.

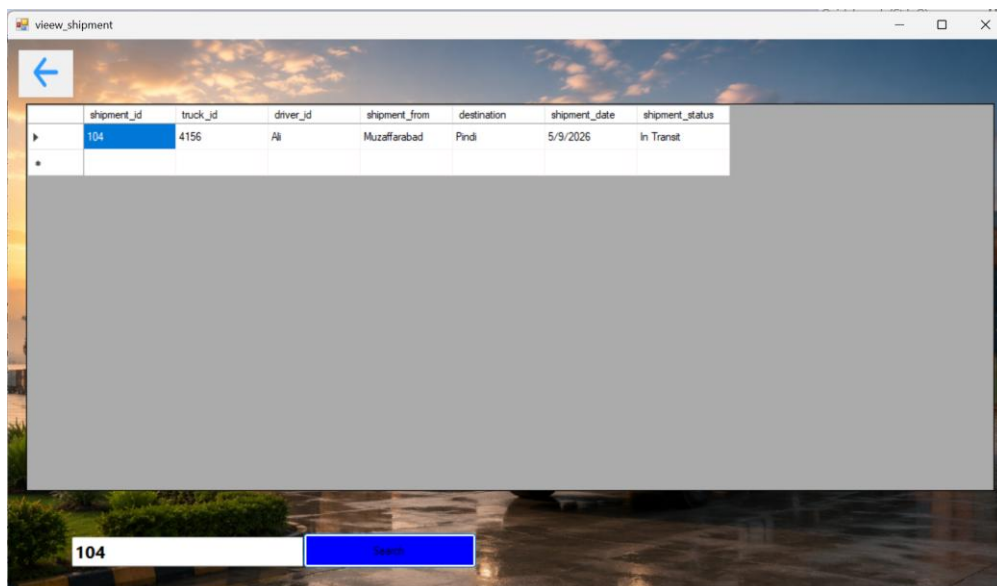


Figure 11.9 — View Shipment After Login

12. Modules Description

The Trucking Management System is organized into four main modules, each handling a specific area of the application:

12.1 Authentication Module

This module handles user login and authentication. It verifies user credentials against the Login table in SQL Server and grants access based on the role (Admin or User). It is the entry point of the entire application.

- Login Form with username and password validation
- Role-based access control (Admin vs User)
- SQL query-based authentication
- Session management after login

12.2 Driver Management Module

This module handles all operations related to drivers. Administrators can add new drivers and view all registered drivers. The data is stored in and retrieved from the Available Driver table.

- Add new driver records to the database
- View all existing driver records in DataGridView
- Driver status management (Available / Assigned)

12.3 Truck Management Module

This module manages all truck-related records. It provides forms to register new trucks and view existing trucks. Truck data is stored in the Available Truck table in SQL Server.

- Register new trucks with all required details
- View all registered trucks in DataGridView
- Truck status tracking (Available / In Use)

12.4 Shipment Management Module

This is the core module of the system. It manages the creation and viewing of shipments. Each shipment links a driver and a truck to an origin-destination pair with a date and status.

- Create new shipment records

- View all shipments in DataGridView
- Search shipments by Shipment ID
- Manage shipment statuses

13. Working Procedure

The following steps describe how the Trucking Management System works from start to finish:

19. Launch the Application: The user runs the application from Visual Studio or the executable file.
20. Login Screen: The Login Form appears. The user enters their username and password.
21. Authentication: The system checks the credentials against the Login table in SQL Server.
22. Access Granted: If login is successful, the user is redirected to the Main Dashboard.
23. Navigate to Module: The user selects a module (Driver / Truck / Shipment) from the Dashboard.
24. Add Records: In the Add Driver or Add Truck forms, the user fills in details and clicks Save. The system executes a SQL INSERT query.
25. View Records: In the View Driver or View Truck forms, all records are loaded using a SQL SELECT query and displayed in a DataGridView.
26. Manage Shipments: The user creates a new shipment by selecting a driver, truck, origin, destination, and date. The record is saved to the Shipment Management table.
27. Search Shipment: The user can search for a specific shipment by entering the Shipment ID in the search box.
28. Logout: The user closes the application when done.

14. Features of the System

The Trucking Management System includes the following key features:

Security Features

- Secure user authentication with username and password
- Role-based access control (Admin / User)
- Password-protected login screen

Driver Management Features

- Add new drivers with complete details
- View all registered drivers in a grid
- Track driver availability status

Truck Management Features

- Register new trucks with registration numbers and capacity
- View all available trucks in a grid
- Track truck status (Available / In Use)

Shipment Management Features

- Create new shipments with origin, destination, driver, and truck
- View all shipments in a DataGridView
- Search shipments by Shipment ID
- Manage shipment status (Pending / In Progress / Delivered)

Database Features

- SQL Server relational database with 4 structured tables
- CRUD operations (Create, Read, Update, Delete)
- ADO.NET database connectivity
- Structured SQL queries for all operations

15. Advantages

- Eliminates manual paperwork and reduces human errors in record-keeping.
- Provides a centralized database for all trucking data — drivers, trucks, and shipments.
- Easy-to-use Windows Forms interface that does not require technical knowledge.
- Fast data retrieval using SQL queries — no need to search through manual files.
- Secure login system ensures that only authorized users can access the data.
- Can handle multiple records efficiently without performance degradation.
- Search functionality allows quick access to specific shipment records by ID.
- Role-based access ensures different levels of access for Admin and regular Users.
- Data is stored permanently in SQL Server — no risk of accidental deletion like paper records.
- The system can be easily extended with more features in future updates.

16. Limitations

- The application is currently limited to desktop use and does not support web or mobile platforms.
- No GPS tracking or real-time location monitoring of trucks.
- The system does not support multiple company branches or locations.
- Report generation (PDF or Excel export) is not implemented.
- No automated email or SMS notifications for shipment updates.
- The application requires SQL Server to be installed on the same machine or network.
- No backup and restore functionality built into the application.
- Limited user roles — only Admin and User are supported.
- No integration with external mapping or route planning services.
- Password storage does not use encryption (plain text in current implementation).

17. Future Enhancements

The following enhancements are planned or suggested for future versions of the Trucking Management System:

- **GPS Integration:** Add real-time GPS tracking for all trucks on an interactive map.
- **Web Application:** Develop a web-based version accessible from any browser without installation.
- **Mobile Application:** Build Android and iOS mobile apps for on-the-go management.
- **Report Generation:** Add the ability to generate PDF and Excel reports for shipments, drivers, and trucks.
- **Email/SMS Notifications:** Automated alerts for shipment status changes.
- **Password Encryption:** Implement SHA-256 or bcrypt encryption for stored passwords.
- **Multi-Branch Support:** Allow multiple company branches to manage their own data.
- **Backup & Restore:** Implement a database backup and restore feature within the app.
- **Dashboard Analytics:** Add charts and graphs to visualize shipment statistics and trends.
- **Cloud Database:** Migrate the database to a cloud platform like Azure SQL for remote access.
- **Document Upload:** Allow uploading of driver license images and truck documents.
- **Billing Module:** Add invoice generation and payment tracking for shipments.

18. Testing

The Trucking Management System was tested manually to ensure all features work correctly. Various test cases were performed on each module to validate the system behavior.

#	Test Case	Input	Expected Output	Result
1	Valid Login	Correct username & password	Redirected to Dashboard	PASS
2	Invalid Login	Wrong password	Error message displayed	PASS
3	Add Driver	All driver details filled	Record saved to DB	PASS
4	Add Driver (Empty)	Empty fields submitted	Validation error shown	PASS
5	View Drivers	Open View Driver form	All records shown in grid	PASS
6	Add Truck	All truck details filled	Truck record saved	PASS
7	View Trucks	Open View Truck form	All trucks shown in grid	PASS
8	Create Shipment	Valid shipment details	Shipment saved to DB	PASS
9	View Shipments	Open shipment view	All shipments displayed	PASS
10	Search Shipment	Valid Shipment ID	Correct record found	PASS
11	Search Shipment	Invalid Shipment ID	No record found message	PASS
12	Database Connect	App launch	Connected to SQL Server	PASS

19. Learning Outcomes

Working on the Trucking Management System project has been a valuable learning experience. The following key skills and concepts were gained during the development of this project:

- **Database Design:** Learned how to design a relational database with multiple tables, primary keys, and relationships using SQL Server.
- **SQL Queries:** Gained practical experience writing SQL SELECT, INSERT, UPDATE, and DELETE queries for real-world applications.
- **C# Programming:** Improved C# coding skills including object-oriented programming concepts, event handling, and working with .NET libraries.
- **Windows Forms Development:** Learned how to design professional GUI applications using Windows Forms controls such as TextBox, Button, DataGridView, ComboBox, and Label.
- **ADO.NET Database Connectivity:** Understood how to connect a C# application to SQL Server using SqlConnection, SqlCommand, and SqlDataReader classes.
- **CRUD Operations:** Implemented Create, Read, Update, and Delete operations in a real project context.
- **Problem Solving:** Developed problem-solving skills by identifying real-world issues in trucking management and designing software solutions.
- **Project Documentation:** Learned how to document a software project professionally including system design, database design, and testing.
- **Visual Studio IDE:** Gained hands-on experience using Visual Studio for code editing, debugging, and building desktop applications.

20. GitHub README — Project Profile

Trucking Management System

A desktop-based Trucking Management System developed using C#, Windows Forms, and SQL Server as a DBMS course project at Riphah International University.

Technologies Used

- Language: C# (.NET Framework)
- IDE: Microsoft Visual Studio
- UI: Windows Forms (WinForms)
- Database: Microsoft SQL Server
- Database Tool: SQL Server Management Studio (SSMS)
- Connectivity: ADO.NET

Features

- Secure Login System (Username & Password Authentication)
- Driver Registration and Management
- Truck Registration and Management
- Shipment Creation and Management
- Search Shipment by ID
- View Records in DataGridView
- Role-based Access (Admin / User)

Screenshots

[Insert Dashboard Screenshot]

[Insert Login Form Screenshot]

[Insert Shipment Management Screenshot]

Installation

29. Clone this repository to your local machine.
30. Open the solution file (.sln) in Visual Studio.
31. Set up SQL Server and run the SQL scripts to create the database.
32. Update the connection string in the application to match your SQL Server instance.
33. Build and run the application in Visual Studio.

Future Improvements

- GPS Integration for real-time truck tracking
- Web and mobile version
- PDF/Excel report generation
- Cloud database migration

21. Conclusion

The Trucking Management System is a fully functional desktop application that successfully demonstrates the practical application of Database Management System (DBMS) concepts. The project was developed using C# programming language with Windows Forms as the user interface framework, connected to a Microsoft SQL Server database for efficient data storage and management.

Throughout this project, the complete software development lifecycle was followed — from problem identification and requirements gathering to system design, database design, implementation, and testing. The application effectively addresses the real-world challenges faced by trucking companies in managing their drivers, trucks, and shipments.

The system provides a secure login mechanism, intuitive forms for data entry, and organized grid views for data display. All CRUD operations are successfully implemented using ADO.NET, allowing seamless communication between the C# front-end and the SQL Server back-end.

This project has been an excellent learning experience that enhanced practical skills in C# programming, SQL Server database design, Windows Forms application development, and software documentation. The knowledge and experience gained from this project will serve as a strong foundation for more advanced software development projects in the future.

In conclusion, the Trucking Management System successfully meets all its stated objectives and provides a solid, expandable platform that can be enhanced with more features in future development cycles. This project stands as a practical showcase of 4th Semester DBMS skills and can be presented as part of an academic and professional portfolio.

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--- End of Documentation ---

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